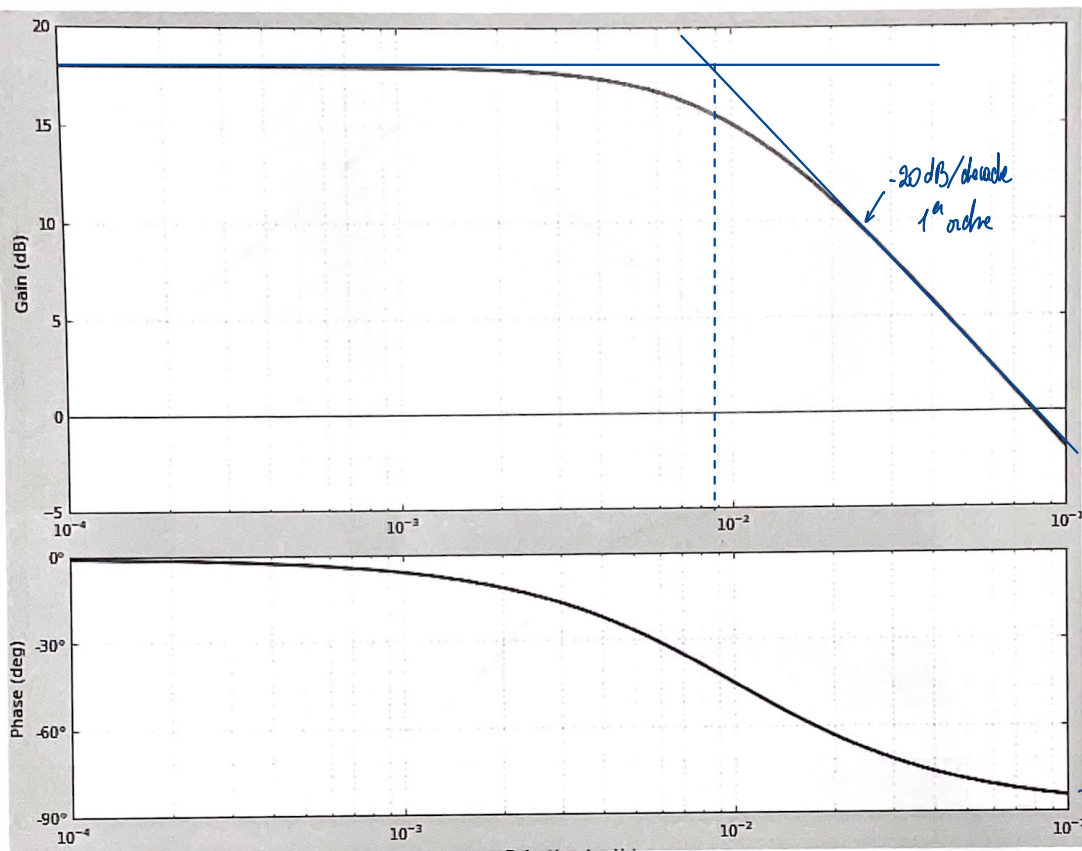
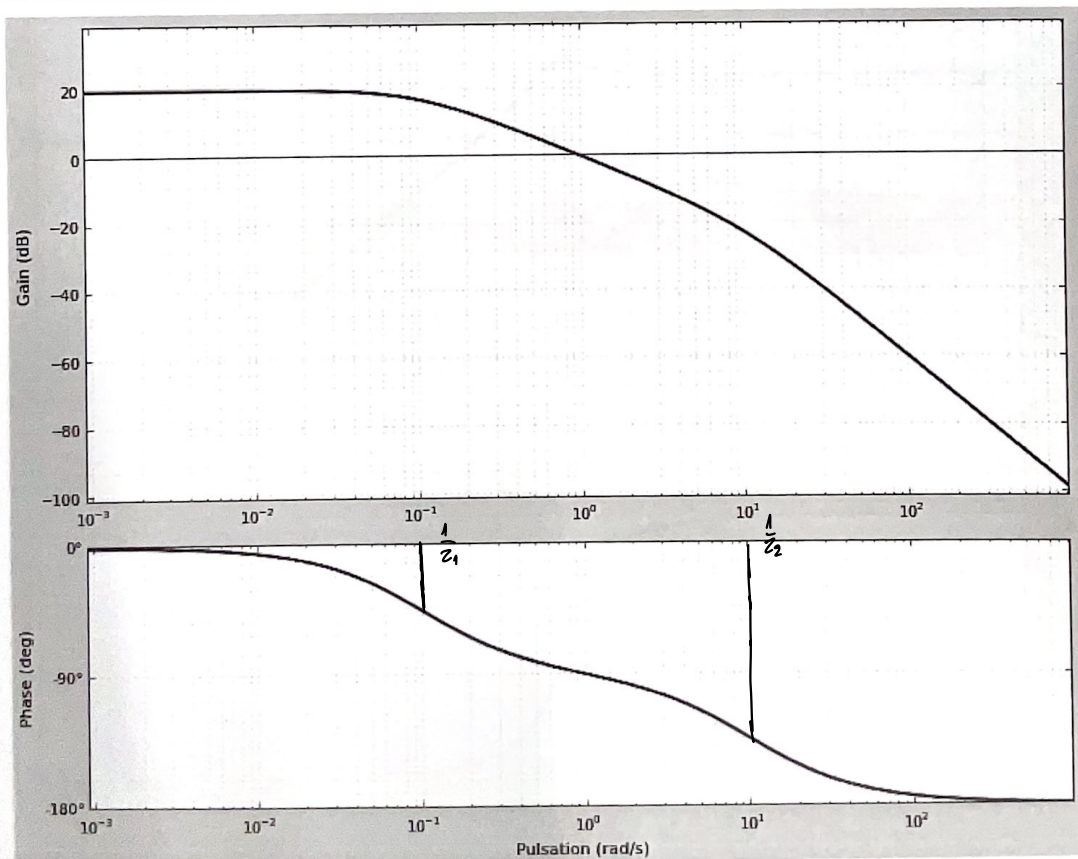


Système 1



Système 2



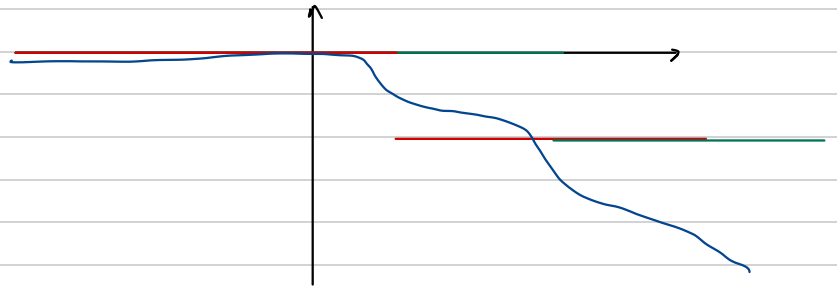
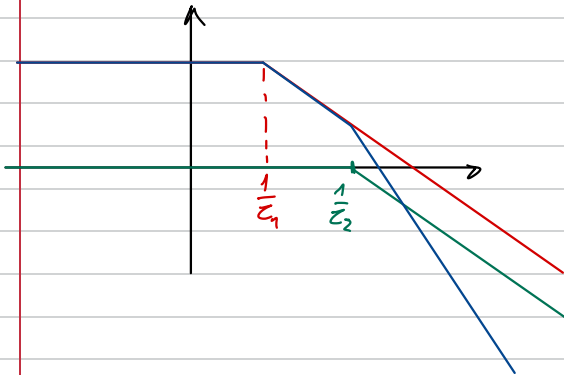
Système 1

$$\frac{1}{\varepsilon} = \omega_c = 10^{-2} \quad \text{Donc } \boxed{\varepsilon = 100\text{s}}$$

$$20 \log K = 18 \text{ dB} \quad \text{Donc } \boxed{K = 10^{\frac{18}{20}} = 7,9}$$

Système 2

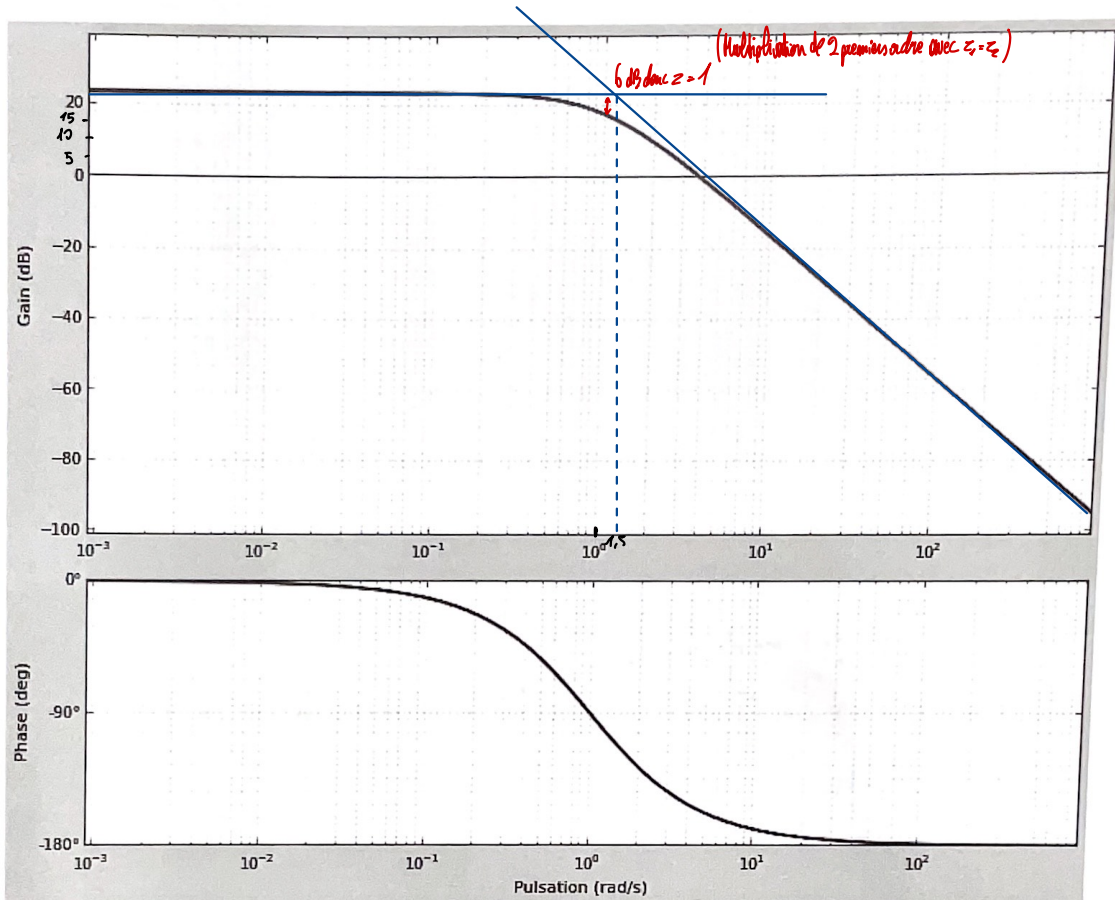
$$\frac{K}{(1 + \varepsilon_1 p)(1 + \varepsilon_2 p)}$$



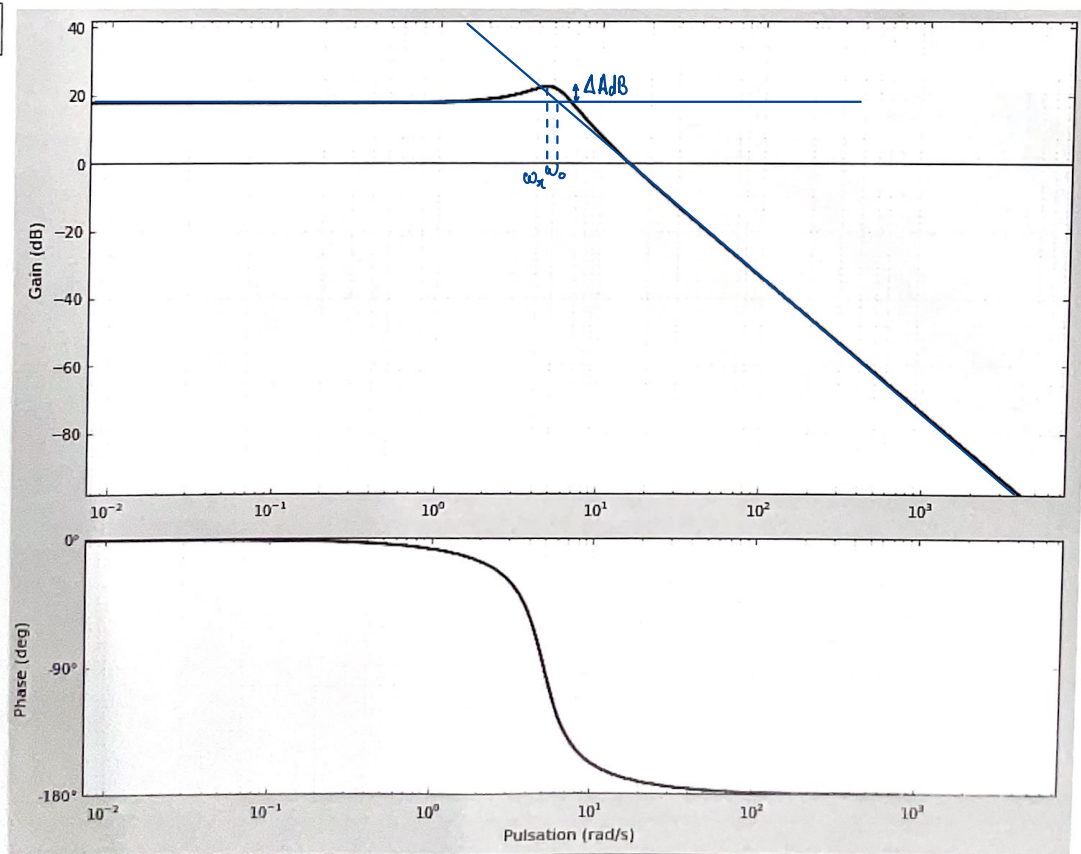
$$\frac{1}{\varepsilon_1} = 10^{-1} \quad \text{Donc } \boxed{\varepsilon_1 = 10\text{s}} \quad \frac{1}{\varepsilon_2} = 10^{-2} \quad \text{Donc } \boxed{\varepsilon_2 = 0,1\text{s}}$$

$$20 \log |K| = 20 \quad \text{donc } \boxed{K = 10}$$

Système 3



Système 4



Système 3:

$$20\log(K) = 22 \quad \text{Donc} \quad K = 10^{\frac{22}{20}} = 12,6 \quad \omega_0 = 1 \text{ rad/s}$$

Pas de résonance $\Rightarrow 0,7$ et pas le même $z < 1$

$$\text{Donc} \quad \boxed{z=1}$$

Système 4:

$$A(\omega) = 20\log\left(\frac{K}{2z\sqrt{1-z^2}}\right)$$

$$= 20\log K - 20\log(2z\sqrt{1-z^2})$$

$$-20\log(2z\sqrt{1-z^2}) = 5$$

$$2z\sqrt{1-z^2} = 10^{-\frac{5}{20}}$$

$$4z^2(1-z^2) = 10^{-\frac{10}{20}} = \frac{1}{\sqrt{10}}$$

$$4z^2 - 4z^4 - \frac{1}{\sqrt{10}} = 0$$

$$Z = z^2 \quad -4Z^2 + 4Z - \frac{1}{\sqrt{10}} = 0$$

$$\Delta = 16 - \frac{4}{\sqrt{10}}$$

$$Z_1 = \frac{4 - \sqrt{\Delta}}{2 \cdot 4}$$

$$Z_2 = \frac{4 + \sqrt{\Delta}}{2 \cdot 4}$$

$$= \frac{4 - \sqrt{10,5}}{8}$$

$$= \frac{4 + \sqrt{10,5}}{8}$$

$$= 0,29$$

$$= 0,91$$

$$z_1 = 0,29$$

$$z_2 = 0,95$$

$$z_1 = 0,29$$

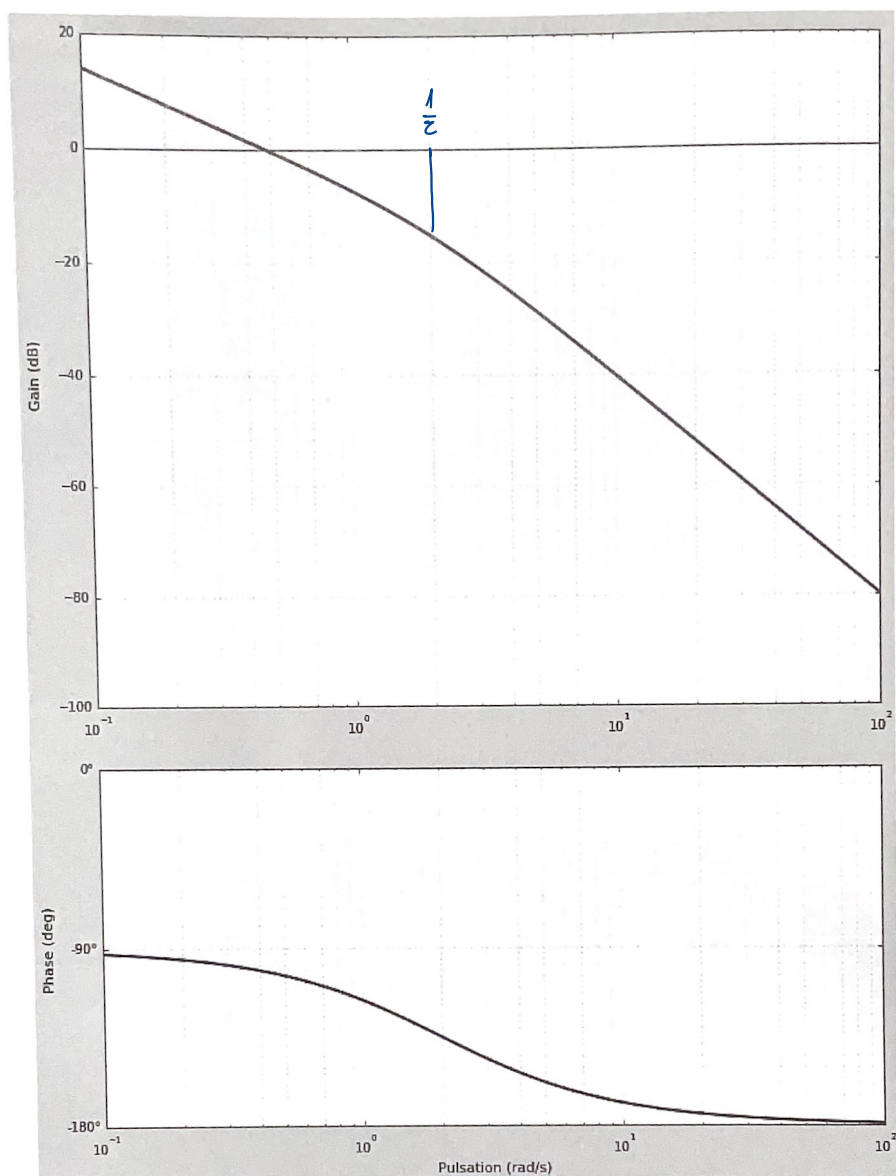
$$z_2 = 0,95$$

$$\text{Donc} \quad \boxed{z=0,29}$$

$$> 0,7$$

pas possible puisque
résonance

Système 5



Système 5:

$$H(p) = \underbrace{\frac{1}{p}}_{\text{intégrateur}} \times \underbrace{\frac{1}{1+\varepsilon p}}_{\text{1er ordre}}$$

$$\frac{1}{z} = 2s \rightarrow z = \frac{1}{2}s = 0,5s \quad \text{et } k=0,5$$